ci sono critiche da neuroscienziati o psicologi cognitivi che riflettano l'idea che l'AI generativa non aiuti a comprendere come funziona il cervello umano perché basata su metodi diversi da quelli del pensare umano?

Generative AI vs. Human Brain Function: Cognitive and Neuroscientific Critiques

No, generative artificial intelligence (specifically Large Language Models) is not considered a valid model for understanding the human brain because its biological and logical operations diverge radically from human thought. This epistemological critique, championed by prominent neuroscientists and cognitive scientists, highlights three fundamental dimensions where AI architectures fundamentally differ from organic cognition.

1. Data Efficiency and Learning (The Brain Learns with Minimality)

Generative AI models are fundamentally data-hungry, whereas the human brain displays extraordinary efficiency.

• **Billions of Words vs. Few Years of Life:** Training a modern LLM requires **trillions of tokens** (essentially the vast majority of digitized human knowledge)—a volume of text a human could not read in a thousand lifetimes.

• **World-Driven Learning:** A young child acquires language and a cohesive mental model of reality using a tiny fraction of that data, relying heavily on physical, visual, and social interaction. AI demonstrates how to *simulate* language by accumulating massive statistics, but it fails to explain how the human brain *generates genuine knowledge* from sparse stimuli.

2. Architecture and Energy Consumption: Two Opposing Worlds

The way AI consumes energy and processes information stands in direct opposition to the evolutionary principles of biological brains.

• **The Energy Factor:** The human brain operates on approximately **20 Watts** of power (the equivalent of a dim lightbulb) while continuously learning and altering synaptic strength via neural plasticity. Conversely, models like GPT-4 demand **industrial-scale datacenters** and megawatts of energy for training.

• **Embodied Cognition:** Cognitive neuroscientists emphasize that human thought is *embodied* (incarnated). The brain evolved to control a physical body within a three-dimensional space, governed by emotions, homeostatic drives, and sensory feedback. Generative AI processes strings of text or pixels in a disembodied vacuum, unable to capture how biological systems formulate meaning.

3. The Fundamental Distinction Between 'Prediction' and 'Understanding'

Generative models rely heavily on the mechanism of **attention and next-token prediction**. The human brain operates via a much more structured, layered causal architecture.

• **Lack of an Internal Causal Model:** AI creates fluent outputs by reshuffling statistical patterns but lacks an internal framework for cause and effect.

• **The Search for the 'Why':** Linguists and cognitive scientists argue that human intellect does not operate via mere statistical probability. The human brain utilizes innate structural frameworks (such as universal grammar) that seek the *explanation of causes*, not just the *prediction of effects*. AI generates plausible answers without understanding 'why', whereas the human brain prioritizes the 'why' even when it cannot

perfectly predict the 'what'.

Summary of Cognitive Differences

Dimension	Human Mind	Generative AI
Information Processing	Rooted in logic, emotion, somatic markers, and conceptual understanding.	Statistical and probabilistic computation over text/data strings.
Cognitive Effort	Engages stable synaptic connections that enhance neural plasticity.	Outsources computational load, bypassing biological neural engagement.
Critical Thought	Forged through doubt, spontaneous error, and constructive disagreement.	Prone to sycophancy bias to maximize user gratification metrics.

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